

DHARANESHWARAN B

Mechanical Engineer Portfolio
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PROFESSIONAL SUMMARY

Final-year Mechanical Engineering student with hands-on experience in CAD modeling, vehicle design, and prototype fabrication. Working knowledge of Siemens NX, SolidWorks, and AutoCAD, with the ability to design and assemble mechanical models, complemented by practical exposure to BAJA vehicle development, engineering documentation, and Modelica-based system simulation. Eager to build a career in Design Engineering, Automotive Systems, and Research & Development, backed by strong analytical, problem-solving, and collaborative skills.

TECHNICAL SKILLS

CAD Software: Siemens NX, SolidWorks, AutoCAD
Engineering: Reverse Engineering, Product Benchmarking, Prototype Development, PLM, Engineering Graphics
Documentation: Technical Documentation, BOM Preparation, MS Office
Programming: Python
Key Roles: Team Leadership, SAEISS Member, ADMMS Conference Volunteer

EXPERIENCE

CAD Designer | ***SAE INDIA – E-BAJA & M-BAJA*** *Nov 2025 – Jan 2026*

- Contributed to securing ₹8 lakh in sponsorship funding from SAE to support BAJA prototype design and fabrication for Team KIOT BAJA.
- Designed and assembled 10+ CAD assemblies for off-road vehicle subsystems using Siemens NX and SolidWorks.
- Produced 15+ fabrication drawings and engineering documentation to support manufacturing and assembly.

Modelica Simulation Trainee | *Modelon* *Dec 2025*

- Built 1D engineering models using Modelica-based simulation tools.
- Developed 2 dynamic mechanical system models, including Spring–Mass–Damper and Tank systems.
- Applied simulation principles to analyses system behavior and engineering performance.

PROJECTS

Loop Frame & Z-Structure Bicycle Design *Oct 2024*

- Designed lightweight bicycle frame concepts using Siemens NX and SolidWorks.
- Improved structural stability through optimized frame geometry.

Portable Power Weeder for Coconut Farm *Mar 2026*

- Engineered and fabricated a portable agricultural power weeder by modifying a brush cutter with a replaceable spiked disc mechanism.
- Improved ease of maintenance and operator safety; created CAD models and conducted performance testing.

Passive Cooling System *Dec 2025*

- Developed a sustainable passive cooling solution without electrical power, applying thermal insulation and airflow principles.
- Fabricated and validated a working prototype.

Autonomous Hospital Supply Delivery Robot *Aug 2026*

- Built an Arduino UNO robot with IR line-following and ultrasonic obstacle detection for autonomous hospital supply delivery.
- Designed and validated a closed-loop proportional control algorithm for navigation and obstacle avoidance.

EDUCATION

B.E. Mechanical Engineering — Knowledge Institute of Technology (KIOT), Salem CGPA: 8.409
Higher Secondary (12th) — Government Boys Higher Secondary School — 68.3% SSLC (10th) — Government Boys Higher Secondary School — Passed

ACHIEVEMENTS

- Contributed to Team KIOT BAJA, achieving All India Rank 1 (AIR 1) and winning the Overall Championship at BAJA SAE INDIA Southern Championship 2026.
- Earned 1st Place in Maneuverability, 1st Place in Suspension, and a total cash prize of ₹1.40 lakh through excellence in vehicle design, testing, and competition performance.
- Best Design Award (₹10,000) – SAE Bicycle Design Challenge.
- 2nd Prize, Overall Performance (₹20,000) – SAE Bicycle Design Challenge.
- 3rd Place – Electric Four-Wheeler Design Challenge.
- Qualified for Regional Round – TN Skill Competition Level-1 (Mechanical Engineering CAD).
- 3rd Place – Mr. Professor Event, Nandha Engineering College.